

L273 FOCUS MOTOR AND PI SCREWING MACHINE

OPERATION MANUAL

MACHINE FRONT PANEL



Figure 1 : Main panel

MAIN PANEL

No	Function	Description
1	AUTO MODE	WHEN TURNED ON, THE MACHINE IS IN AUTO MODE AND READY FOR PRODUCTION. WHEN TURNED OFF, THE MACHINE IS IN DEBUGGING OR REPAIRING MODE.
2	SEMI AUTO MODE	TO ENTER SEMI AUTO MODE
3	MANUAL	TO ENTER IO & SERVO SETTING MODE
4	SETTING	TO ENTER SETTING MODE
5	CALIBRATION AND TESTING	TO ENTER DRIVER CALIBRATION MODE
6	ALARM HISTORY	TO CHECK ALARM HISTORY

MACHINE OPERATION PANEL

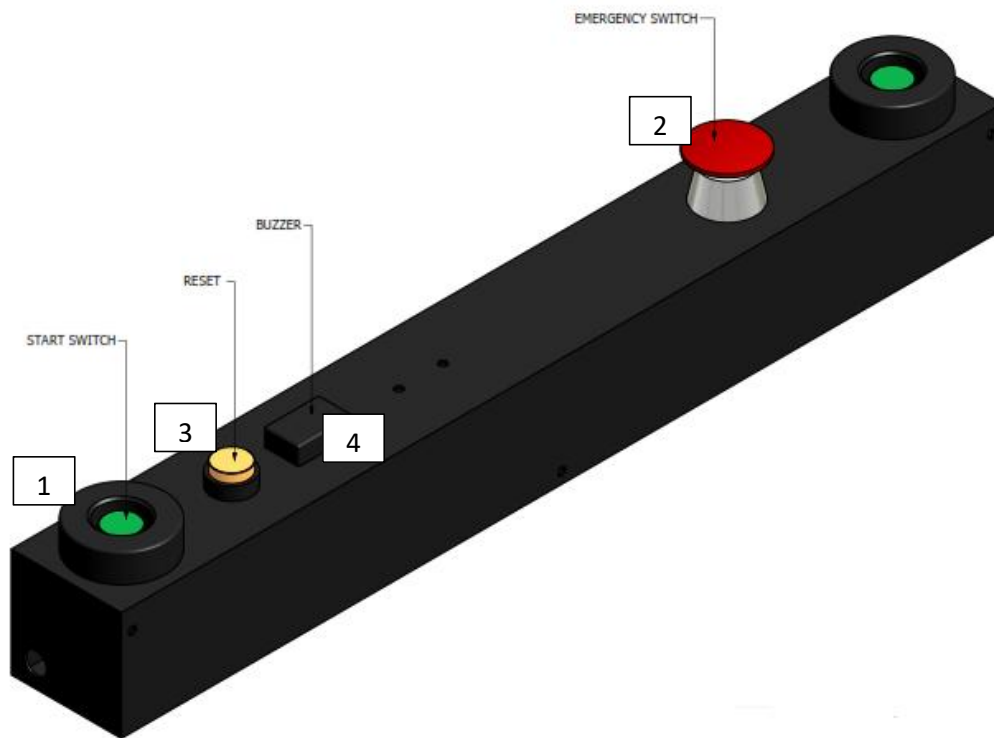


Figure 2 : Operation panel

OPERATION PANEL

No	Function	Description
1	START BUTTON	PRESS TO START THE MACHINE
2	EMERGENCY BUTTON	PRESS TO EMERGENCY STOP
3	RESET BUTTON	PRESS TO RESET THE MACHINE AND BUZZER
4	BUZZER	WILL TURN ON WHEN PROCESS COMPLETE OR EMERGENCY STOP

MACHINE OPERATION PROCEDURE

1. POWER ON THE MACHINE BY SWITCH ON POWER BUTTON.
2. TURN ON AIR SUPPLY TO THE MACHINE AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.
3. PRESS RESET BUTTON AND MACHINE WILL PERFORM HOMING ROUTINE.
4. GOTO SETTING MODE AND SELECT SLIDER SPEED.
5. SET MACHINE TO AUTO MODE.
6. LOAD PRODUCT IN MACHINE AND PRESS THE START BUTTON.
7. WHEN PROCESS IS COMPLETED, THE BUZZER WILL SOUND 3 TIMES AND OPERATOR CAN UNLOAD THE PRODUCT FROM THE MACHINE.
8. REPEAT PROCESS 5 AND 6 TO CONTINUE PRODUCTION.

SETTING FUNCTION

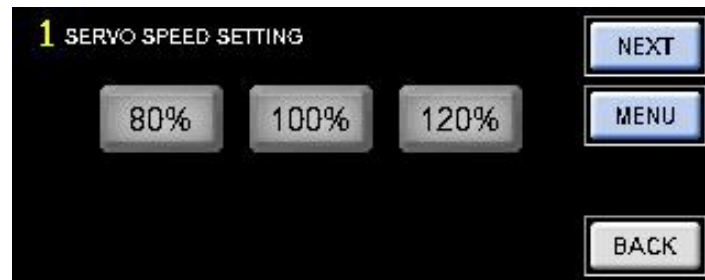


Figure 3 : Speed Screen

SPEED SETTING SCREEN

No	Buttons	Description
1	120%	Machine run at 120% of normal speed
2	100%	Machine run at 100% of normal speed
3	80%	Machine run at 80% of normal speed
4	BACK	Back to previous page
5	NEXT	Go to next page

10



Figure 4 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	KORO	To start the koro sequence
2	SCREW	To start the screwing sequence by selecting screwing number(1,2,3,4,6,7,8, & 9)
3	GLUE	To start the glue sequence by selecting glue number (1 & 2)
4	UV	To start the UV sequence



Figure 5 : Timing Screen

TIMING SCREEN

No	Function	Description
1	PICK TIMEOUT	Timeout for screw & koro pick
2	RETRY	Retry for screw & koro pick
3	TIMEOUT	Timeout for screw finish
4	SLOW PULSE	Speed change for Z Axis point
5	SCREW WAIT	Screw pickup wait
6	GLUE SPEED	Servo speed while GLUE



Figure 6 : Servo Speed Screen

SERVO SPEED SCREEN

No	Function	Description
1	ACC	Acc speed for each of the axis
2	DEC	Dec speed for each of the axis
2	OPERATING	Operating speed for each of the axis



Figure 7 : Other servo Speed Screen

OTHER SERVO SPEED SCREEN

No	Function	Description
1	BTM ROTATE	Bottom rotate servo Acc , Decc ,& Operating speed
2	GLUE AXIS	Glue axis servo Acc , Decc ,& Operating speed
2	SIDE ROTATE	Side rotate servo Acc , Decc ,& Operating speed

SEMI AUTO FUNCTION



Figure 8 : Semi Auto Screen

SEMI AUTO SCREEN

No	Function	Description
1	SCREW 1	Screw 1 process will be turned on when selected
2	SCREW 2	Screw 2 process will be turned on when selected
3	SCREW 3	Screw 3 process will be turned on when selected
4	SCREW 4	Screw 4 process will be turned on when selected
5	SCREW 5	Screw 5 process will be turned on when selected
6	SCREW 6	Screw 6 process will be turned on when selected
7	SCREW 7	Screw 7 process will be turned on when selected
8	SCREW 8	Screw 8 process will be turned on when selected
9	SCREW 9	Screw 9 process will be turned on when selected

10	GLUE 1	Apply Glue process 1 will be turned on when selected
11	GLUE 2	Apply Glue process 2 will be turned on when selected
12	UV	UV process will be turned on when selected

MANUAL FUNCTION



Figure 9 : Semi Auto Screen

MANUAL SCREEN

No	Function	Description
1	I/O CONTROL	Go to I/O Control screen
2	SERVO CONTROL	Go to Servo Control screen

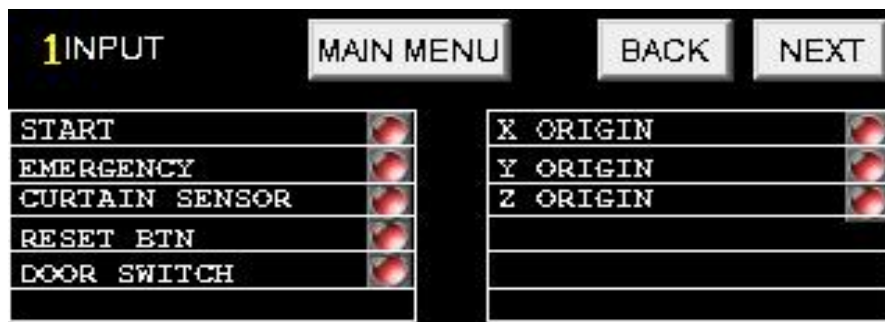


Figure 10 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals

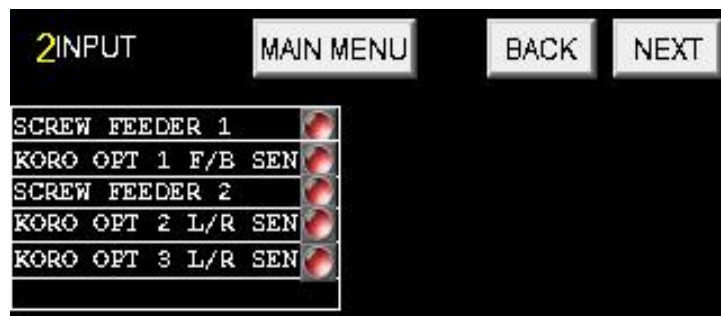


Figure 11 : Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	Show status of the input signals

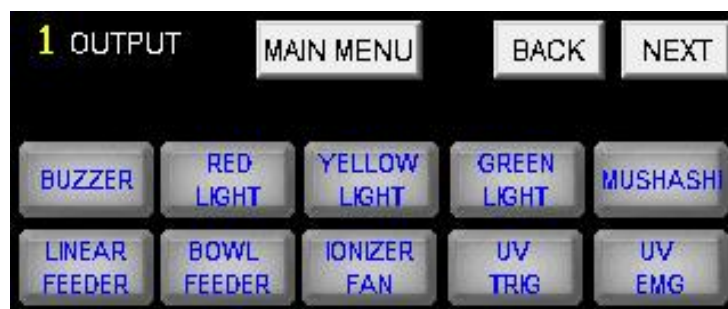


Figure 12 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

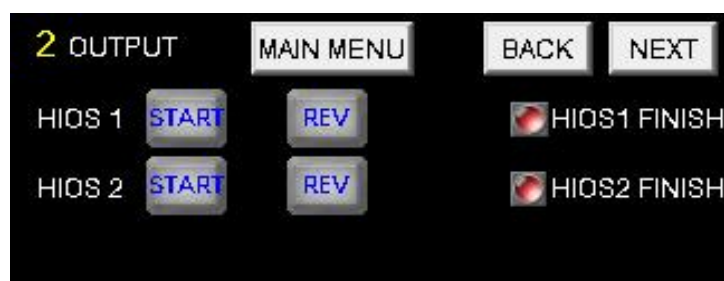


Figure 13 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

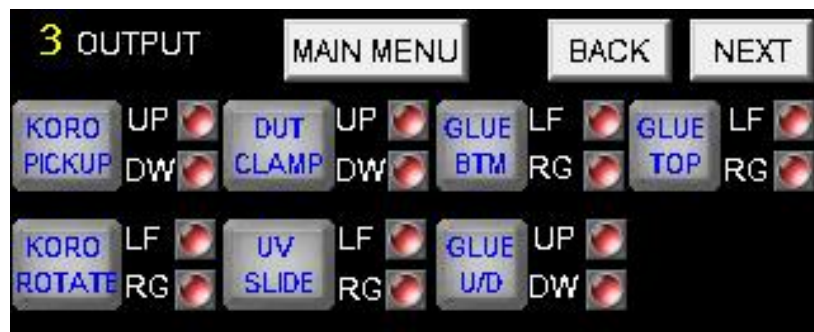


Figure 14 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

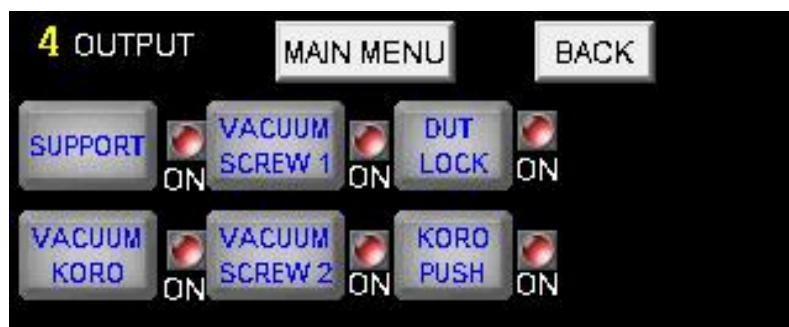


Figure 15 : Output Screen

OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Menu to trigger all output signals

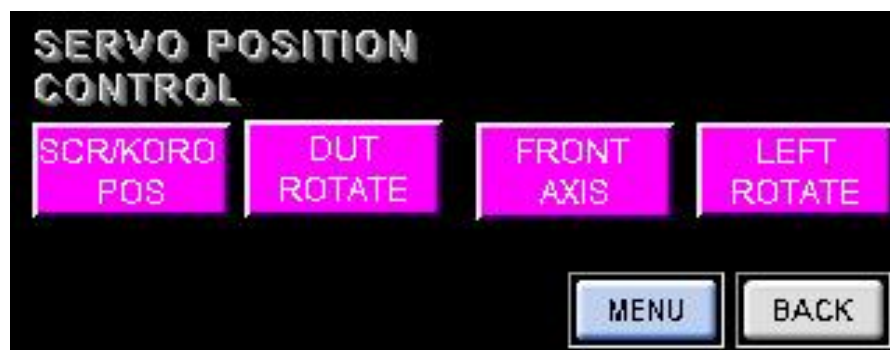


Figure 16 : Servo Control Menu Screen

SERVO CONTROL SCREEN

No	Function	Description
1	SCR/KORO POS	Go to screen to teach screw and koro pick & place position
2	DUT ROTATE	Go to screen to teach base pick up position
3	FRONT AXIS	Go to screen to teach tray drop position
4	LEFT ROTATE	Go to screen to teach part carrier slide position



Figure 17 : Top Tray Position Screen

TOP TRAY SCREEN

No	Function	Description
1	X+	Move X-Axis to the right
2	X -	Move X-Axis to the left
3	Y+	Move Y-Axis to the frontward
4	Y-	Move Y-Axis to the backward
5	Z+	Move Z-Axis to the downward
6	Z-	Move Z-Axis to the upward
7	Z UP	Move Z-Axis to Z origin point
8	Z DOWN	Move Z-Axis to selected point no.
9	DOWN SAVE	Save Z-Axis to selected point no.
10	SLOW	Make jog slow or fast
11	VAC KOR	Turn on vacuum
12	KORO PICKUP	Up/Down Koro cylinder
13	KORO ROT	Left/Right Koro rotate
14	SCR VAC	Turn on screw vacuum
15	GOTO	Move X & Y Axis to selected point no.
16	HOME	Move X,Y & Z Axis to Origin position
17	SAVE	Save X-Axis & Y-Axis to selected point no.



Figure 18 : DUT Rotate Position Screen

DUT ROTATE SCREEN

No	Function	Description
1	CCW	Move Servo Counter-clock-wise
2	CW	Move Servo Clock-wise
3	DUT CLAMP	Turn on Dut clamp
4	SUPPORT	Turn on Support cylinder
5	GOTO	Move X & Y Axis to selected point no.
6	HOME	Move X,Y & Z Axis to Origin position
7	SAVE	Save X-Axis & Y-Axis to selected point no.



Figure 19 :Front axis Position Screen

FRONT AXIS SCREEN

No	Function	Description
1	BACK	Move Rotate Axis Back
2	FRONT	Move Rotate Axis Front
3	GLUE TOP	Turn on Glue Top Cylinder
4	Glue U/D	Up/Down Glue Cylinder
5	UV SLIDE	Left/Right UV Slide
6	GLUE BTM	Turn on Glue Cylinder

7	DUT CLAMP	Turn on DUT clamber
8	GOTO	Move Rotate Axis to selected point no.
9	HOME	Move Rotate Axis to Origin position
10	SAVE	Save Rotate Axis to selected point no.

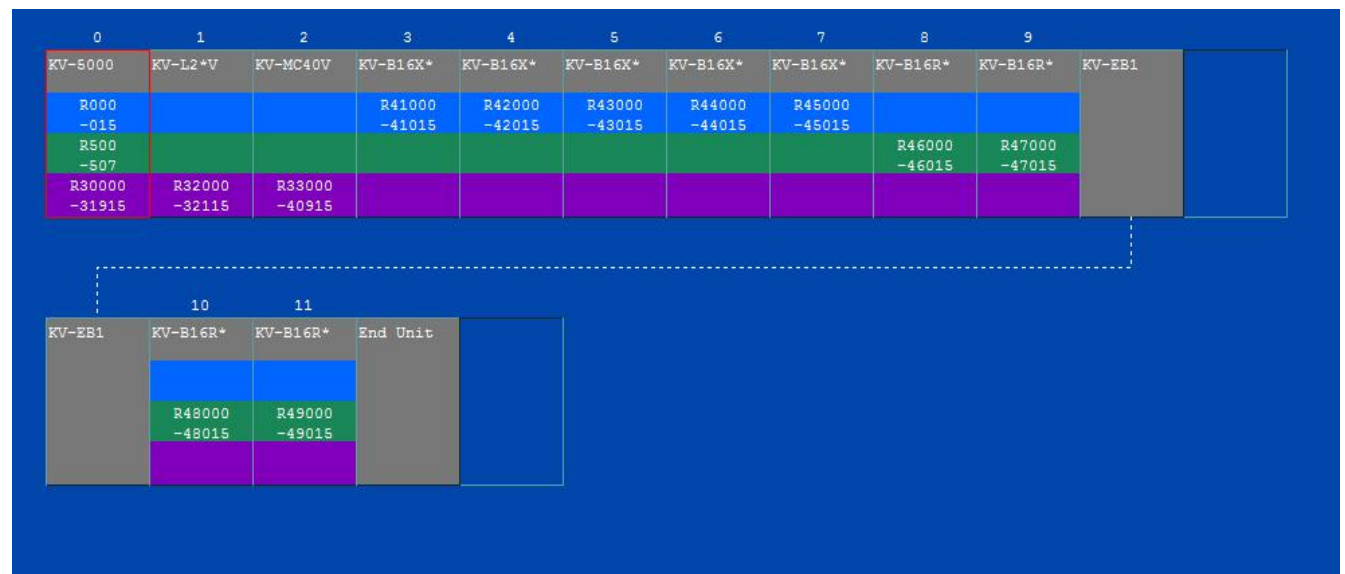


Figure 20 :Side Rotate Position Screen

SIDE ROTATE SCREEN

No	Function	Description
1	CCW	Move Servo Counter-clock-wise
2	CW	Move Servo Clock-wise
3	LOCK	Turn on Lock
4	GOTO	Move Part Carrier Axis to selected point no.
5	HOME	Move Part Carrier Axis to Origin position
6	SAVE	Save Part Carrier Axis to selected point no.

MACHINE PLC SETTING & LADDER DIAGRAM



Base	
Leading DM No.	DM10000
Number of DMs used	16
Leading relay No. (ch unit setting)	R30000
Number of relays used	320
Baud rate	100/10 Mbps, Auto(*)
Setting method of IP address	Fixed IP address(*)
IP address	192.168.0.10
Subnet mask	255.255.255.0
Default gateway	0.0.0.0
Port No. (KVS, KV COM+, DB)	8500
Port No. (host link)	8501
Reception timeout[s].	10
Keep Alive[s]	600
FTP server enabled	disabled(*)
Password	[Unset]
FTP server RUN/PROG switching	Yes(*)
Routing setting	no control(*)
Port No. (VT)	8502

Socket0	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
Socket1	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)
Socket2	
KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Port No. (System expansion)	8504
DNS Server	0.0.0.0
MC protocol port No. (TCP)	5000
MC protocol Port No. (UDP)	5000
MC protocol code	BIN(*)
Type name code of MC protocol	0
Type name of MC protocol	
Simple PLC link port No. (UDP)	0
Auto adjust clock	Not used(*)
SNTP communication timeout[ms]	60
Clock adjustment time time[h]	0
Clock adjustment time[m]	0
Clock adjustment time[s]	0
Clock adjustment interval[m]	60
GMT offset	+(*)
GMT offset[h]	9
GMT offset[m]	0
GMT offset[s]	0
NTP(SNTP) server	0.0.0.0

Socket3

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket4

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket5

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)

Socket0

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket1

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)
Response sub-header	E0
Communication direction	Send(*)

Socket2

KV Socket	disabled(*)
Byte replacement	H->L(*)
Command sub-header	60
Response	Without(*)

Response	Without (*)
Response sub-header	E0
Communication direction	Send (*)
Socket6	
KV Socket	disabled (*)
Byte replacement	H->L (*)
Command sub-header	60
Response	Without (*)
Response sub-header	E0
Communication direction	Send (*)
Socket7	
KV Socket	disabled (*)
Byte replacement	H->L (*)
Command sub-header	60
Response	Without (*)
Response sub-header	E0
Communication direction	Send (*)
Common KV socket	
Response timeout[s]	30

0	1	2	3	4	5	6	
KV-5000	KV-L2*V	KV-MC40V	KV-B16X*	KV-B16X*	KV-B16X*	KV-B16X*	KV-B
R000 -015			R41000 -41015	R42000 -42015	R43000 -43015	R44000 -44015	R4
R500 -507							
R30000 -31915	R32000 -32115	R33000 -40915					

Base

Leading DM No.

DM10100

Number of DMs in use

1

Leading relay No.(set channel unit)

R32000

Number of relays in use

32

Node No.

0(*)

Port 1

Operation mode

KV BUILDER/KV STUDIO mode

Detail

---(*)

Interface

RS-232C(*)

Baudrate

Auto

Data bit length

8 bits(*)

Start bit

1 bits(*)

Stop bit

1 bits(*)

Parity

Even(*)

Checksum

none(*)

RS/CS flow control

OFF(*)

Port 2

Operation mode

KV mode (host link) (*)

Baudrate	Auto
Data bit length	8 bits(*)
Start bit	1 bits(*)
Stop bit	1 bits(*)
Parity	Even(*)
Checksum	none(*)
RS/CS flow control	OFF(*)

Port 2

Operation mode	KV mode (host link) (*)
Detail	--- (*)
Interface	RS-422A/485 (4-wire) (*)
Baudrate	9600bps (*)
Data bit length	8 bits(*)
Start bit	1 bits(*)
Stop bit	1 bits(*)
Parity	Even(*)
Checksum	none(*)

Details setting

Transfer timeout time (secs)	3
------------------------------	---

0	1	2	3	4	5	6	
KV-5000	KV-L2*V	KV-MC40V	KV-B16X*	KV-B16X*	KV-B16X*	KV-B16X*	KV-B16X*
R000 -015			R41000 -41015	R42000 -42015	R43000 -43015	R44000 -44015	R45000 -45015
R500 -507							
R30000 -31915	R32000 -32115	R33000 -40915					

Basic

Leading DM No.	DM10200
Number of occupied DMs	582
Leading relay No. (channel-wise set)	R33000
Number of occupied relays	1280
Function expansion unit	None(*)
Refresh mode	Simple mode(*)

Introduction

This section describes the features and specifications of the Focus Motor And Pi Screwing Machine.

About Focus Motor And Pi Screwing Machine

-The purpose of the system is

- To assembly group 4 unit to group 5 unit by screwing at 4 screw positions.

- Inserting Collar Adjust at 1 position.

- Screwing Focus Motor unit at 3 screw positions.

- Screwing PI at 2 screw positions.

- Apply glue for Hall Generator at 2 bond positions.

- Simple operation

The system is designed and emphasizes on simple, easy teaching and automatic operations.

- Modern and user-friendly user interface

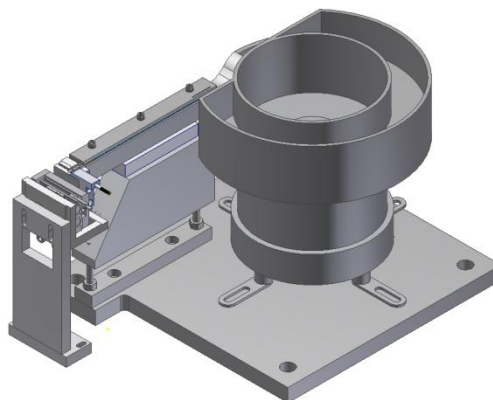
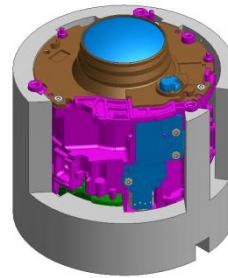
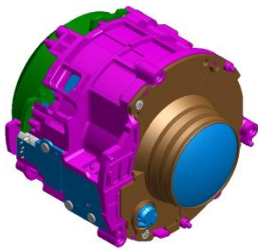
Simple and modern GUI of the controlling software features many functions and even easy to use.

System Specifications

The system specification will be listed below

Mechanical and Electrical Specifications

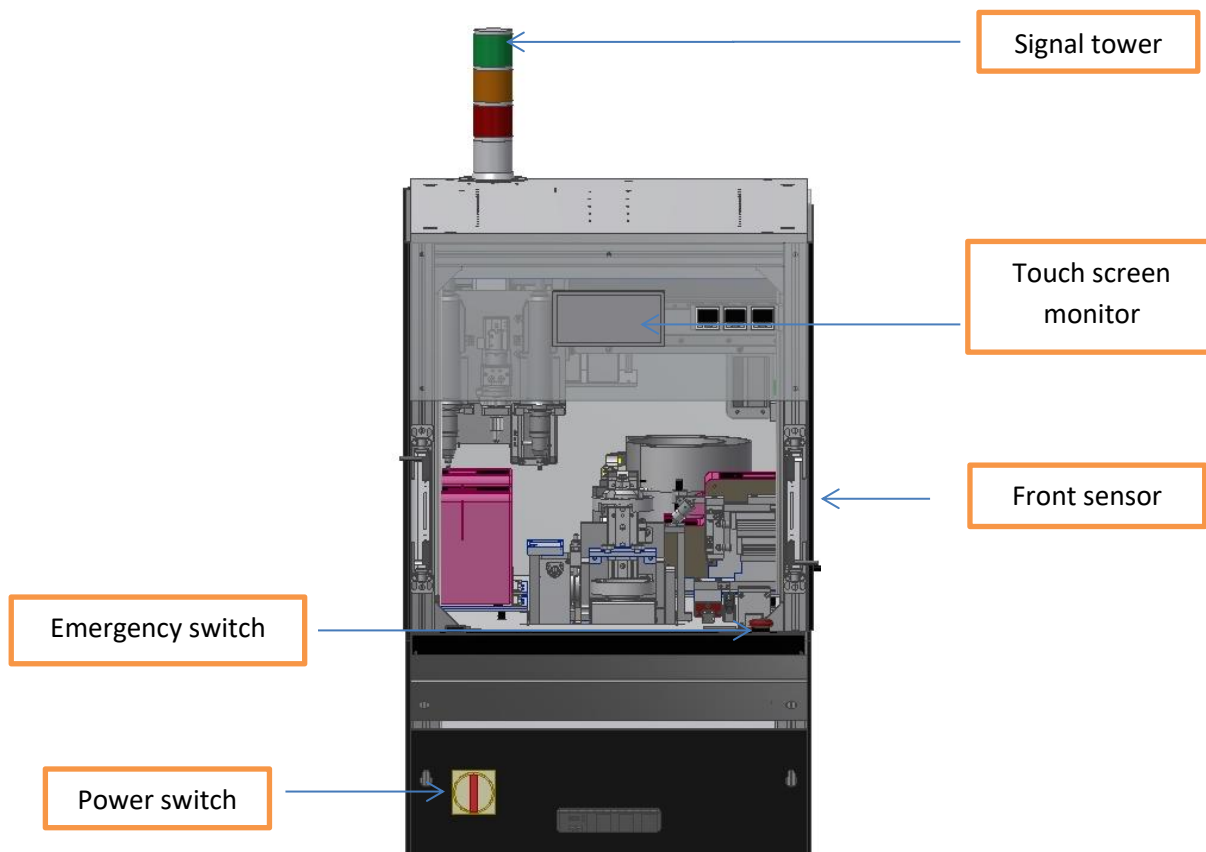
Item	Specifications
Control PLC	Keyence PLC
Power supply	100V AV - 110V AC
Air pressure	Min 0.5 Mpa or 5 bar (dry air)
Cycle time/pcs	49s/pcs



Names and functions of components

This section main describes names and the functions of the various parts of Is Auto Coil Machine.

External View



- Emergency switch

By pressing the emergency switch will bring the machine to emergency halt status.

- Monitor and touch screen

The touch screen monitor is used for machine operations control.

- Front curtain sensor

It is a safety prevention tool, only enabled when performing sorting operation. The operation will be halted if the sensor is **ON**.

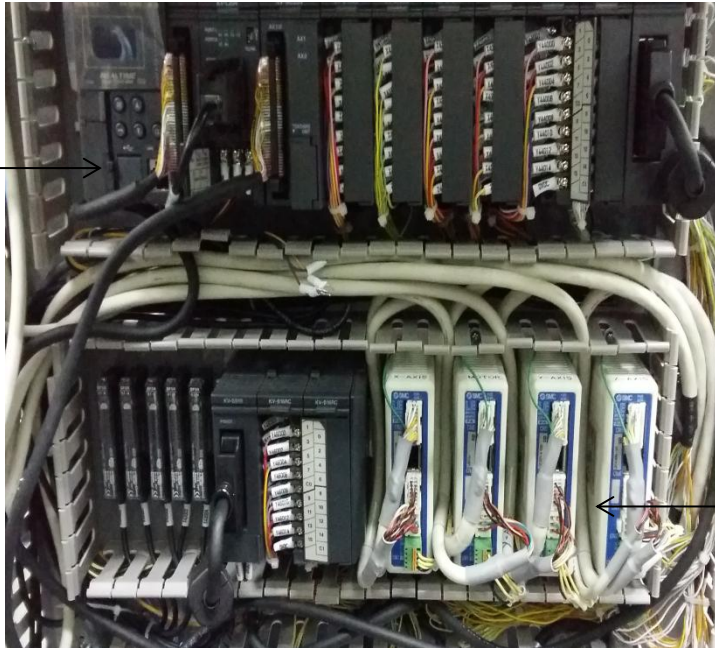
- Signal tower

It shows the system current status with the green, red and yellow lights with buzzer.

Inside Door

Power Box

Keyence PLC
KV 5000

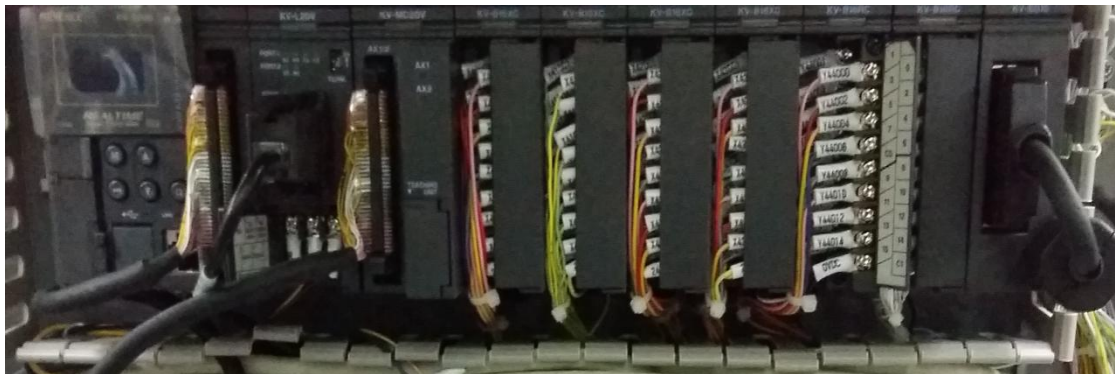


SMC Driver Motor
LE

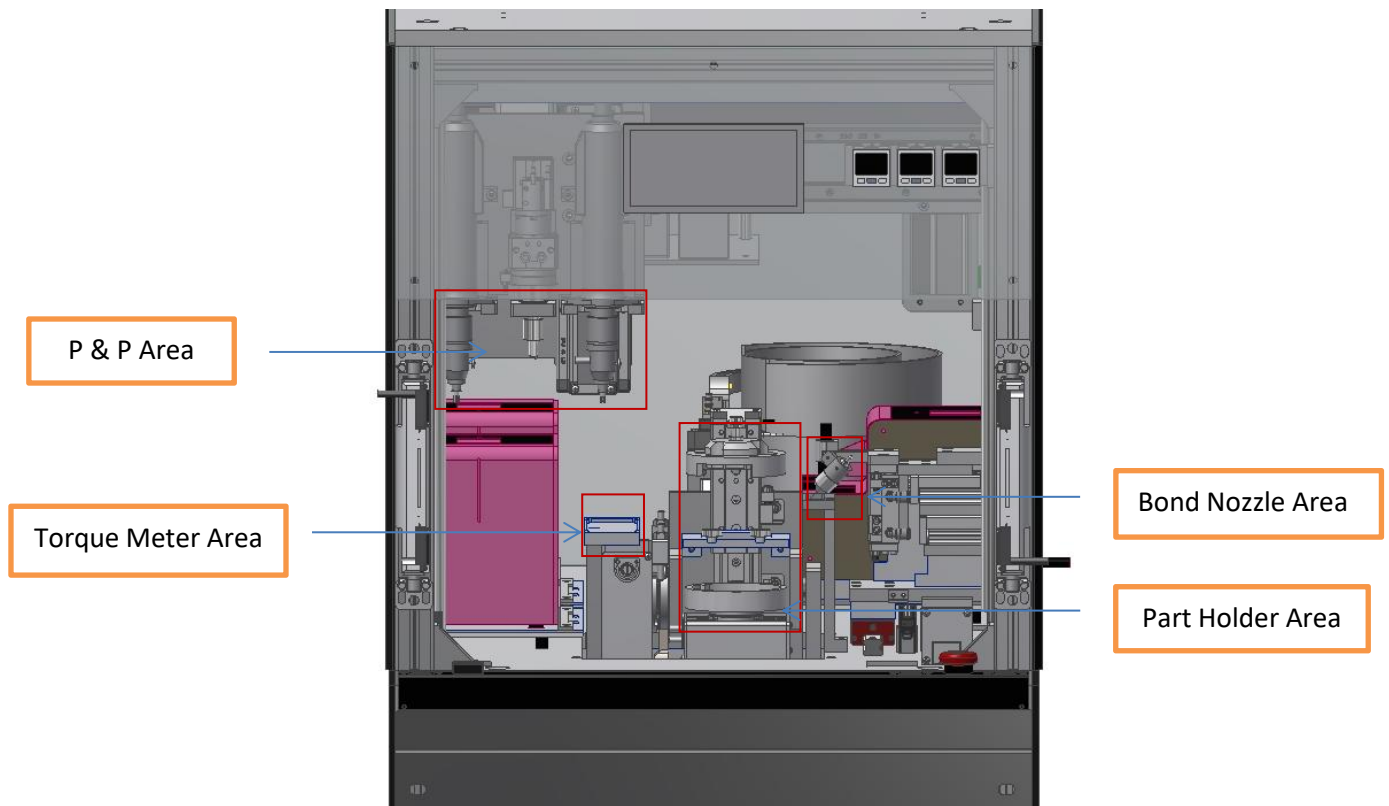
- SMC Servo Motor LE



- Keyence PLC KV 5000



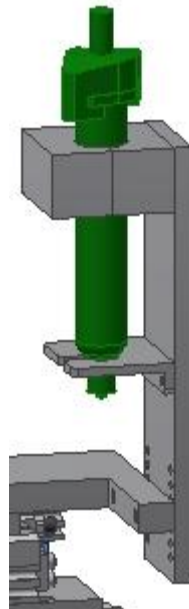
Working Section



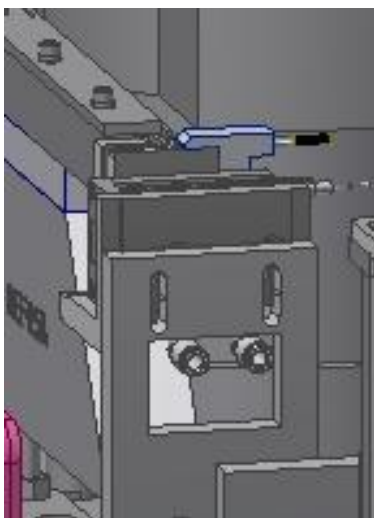
UV Curing Area



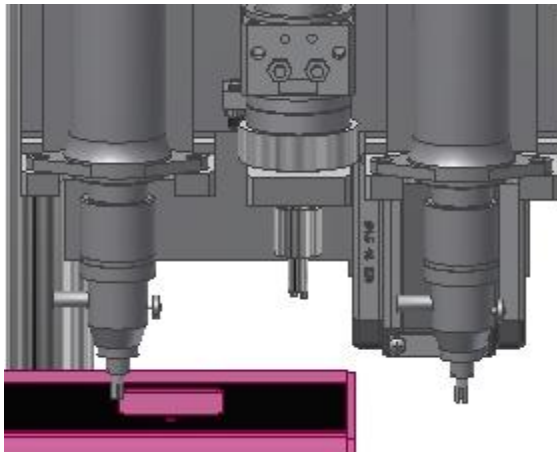
Bond Barrel Area



Collar Pickup Area



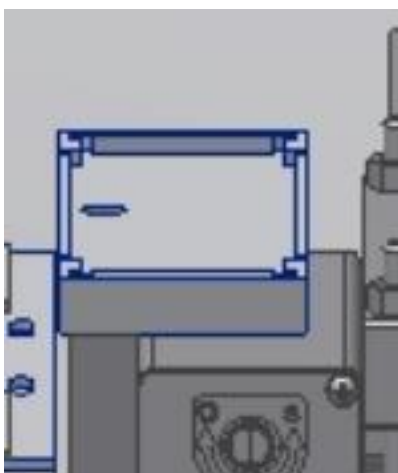
Pick And Place Area



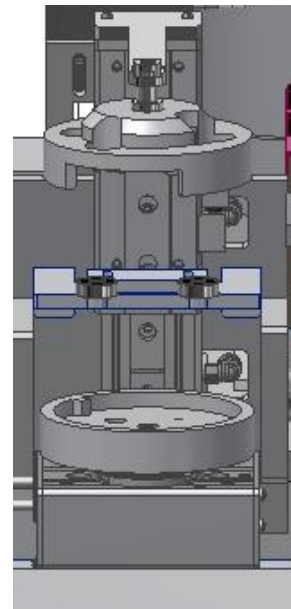
Bond Nozzle Area



Torque Meter Area



Part Holder Area



OPERATION MANUAL

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INTRODUCTION

TOUCH SCREEN PANEL

WIRING DIAGRAM

PART LIST

MACHINE OPERATION PANEL

L273 FOCUS MOTOR **AND PI SCREWING MACHINE**



OPERATION MANUAL